User Study for Guided Active Learning in Recommender Systems

1. Goal (To what end?)

We want to test whether **showing users how their ratings affect future recommendations** (guided active learning) leads to:

- Better recommendation quality,
- Higher user trust and satisfaction,
  compared to a standard system that just asks for ratings without previews.

2. What are we testing? (Variables)

- **Independent variable (condition):**
  - *Standard AL* (user rates movies, no preview).
  - *Guided AL* (user rates movies, sees "if 1★ vs 5★ → how top 10 movies and categories Change").
- **Dependent variables (measures):**
  - **Objective:**
    - Precision : how many recommended movies the user says they'd actually watch.

3. Context

- **Format:** Online web app .
- **Task:** Interactively rate movies (0.5–5 stars). Each time, either with or without preview.
- **Duration:** 10-15 minutes total.

5. Procedure (Step by step)

1. **Introduction & consent**
   a. Participants are welcomed, told the study is about comparing two versions of a movie recommender.
   b. They are reminded that the system is being tested, not them.
   c. They check the "consent" box.
2. **Condition 1 (either Standard AL or Guided AL, randomized)**

a. The system presents 10 movies chosen by an active learning strategy:
   - Uses **matrix factorization (SVD)** to represent movies and users in a latent factor space.
   - **Pre-computes movie embeddings** from the historical MovieLens dataset.
   - **Infers the user vector dynamically** as new ratings are given.
   - Applies **uncertainty sampling**: selects movies whose predicted ratings are closest to the midpoint (2.5).
   - Goal: **ask about the most uncertain movies** to gain the most information about the user's preferences.
b. Participants rate each movie (0.5–5 stars).
c. **If in Guided AL:** next to the rating input, they also see a preview of how their choice would affect the recommendation list ("If 5★ → more sci-fi; If 1★ → more comedies").
d. At the end of the 10 ratings:
   i. The system generates a personalized **Top-10 recommendation list**.
   ii. Participant marks which ones they would actually watch or whether they liked it or not if they watched it .

3. **Short break**

4. **Condition 2 (the other version, with a new set of movies)**
   a. The participant now interacts with the **other version of the system** (if they first did Standard, now they do Guided; if they first did Guided, now Standard).
   b. The system again selects 10 movies from a **different pool** (so they don't just re-rate the same titles).
   c. Process:
      i. Rate movies one by one.
      ii. Depending on the condition: with or without previews of rating impact.
   d. At the end of this block:
      i. A new **Top-10 recommendation list** is generated.
      ii. Participant again marks which movies they would watch.

Why disjoint sets of movies?

e. To avoid bias: if they saw *The Matrix* in both conditions, their second rating might just repeat the first.

f. By giving them **different but comparable movies** in each condition, we ensure the results reflect the **system design**, not memory of past ratings.

7. Analysis

- Compare the Precision@10 between the Guided and Standard conditions, i.e., test whether Guided Active Learning provides more preferred recommendations to participants than Standard Active Learning